

Some project details

Introduction

Background

- COVID-19 is caused by the Severe Acute Respiratory Syndrome CoronaVirus 2
- Approximately 704,753,890 cases worldwide and 7,010,681 deaths worldwide

Motivation

- Explore non-invasive methods through audio analysis (cough sounds) for early COVID-19 detection
- Develop low-cost, accessible diagnostic tools that can be used remotely through smartphones or other devices
- RT-PCR is the golden standard for detecting COVID-19, but it requires specialised equipment, costs money, and is not accessible to everyone

Literature Review

Hamdi et al. (2022)

- Utilized the COUGHVID dataset to identify COVID-19 from healthy patients: attention-based hybrid CNN-LSTM
 - 0.91 accuracy, 0.91 AUC, 0.91 sensitivity

Lella & Pja (2022)

- Classify COVID-19 vs asthma, bronchitis, etc. from cough, voice, breath: multi-channel CNN
 - 0.95 accuracy, 0.97 F1 score

Imran et al. (2020)

- Real-time COVID-19 screening via mobile app: CNN + 3 classifiers (DTL-MC, CML-MC, DTL-BC)
 - 0.92 accuracy, 0.89 sensitivity

Schuller et al. (2021)

- Establish baselines for automatic COVID-19 detection using audio recordings: CNN, LSTM, SCM
 - 74% unweighted average recall

Dataset Description

The COUGHVID crowdsourcing dataset

- Over **25,000** crowdsourced cough recordings representing a wide range of participant ages, genders, geographic locations, and COVID-19 statuses.
- Recordings were collected between **April 1st, 2020** and **December 1st, 2020** through a web application
- Samples labelled as COVID-19 originate from countries where the virus was prevalent at the time of recording, and an evaluation of the quality of the recordings containing cough sounds.

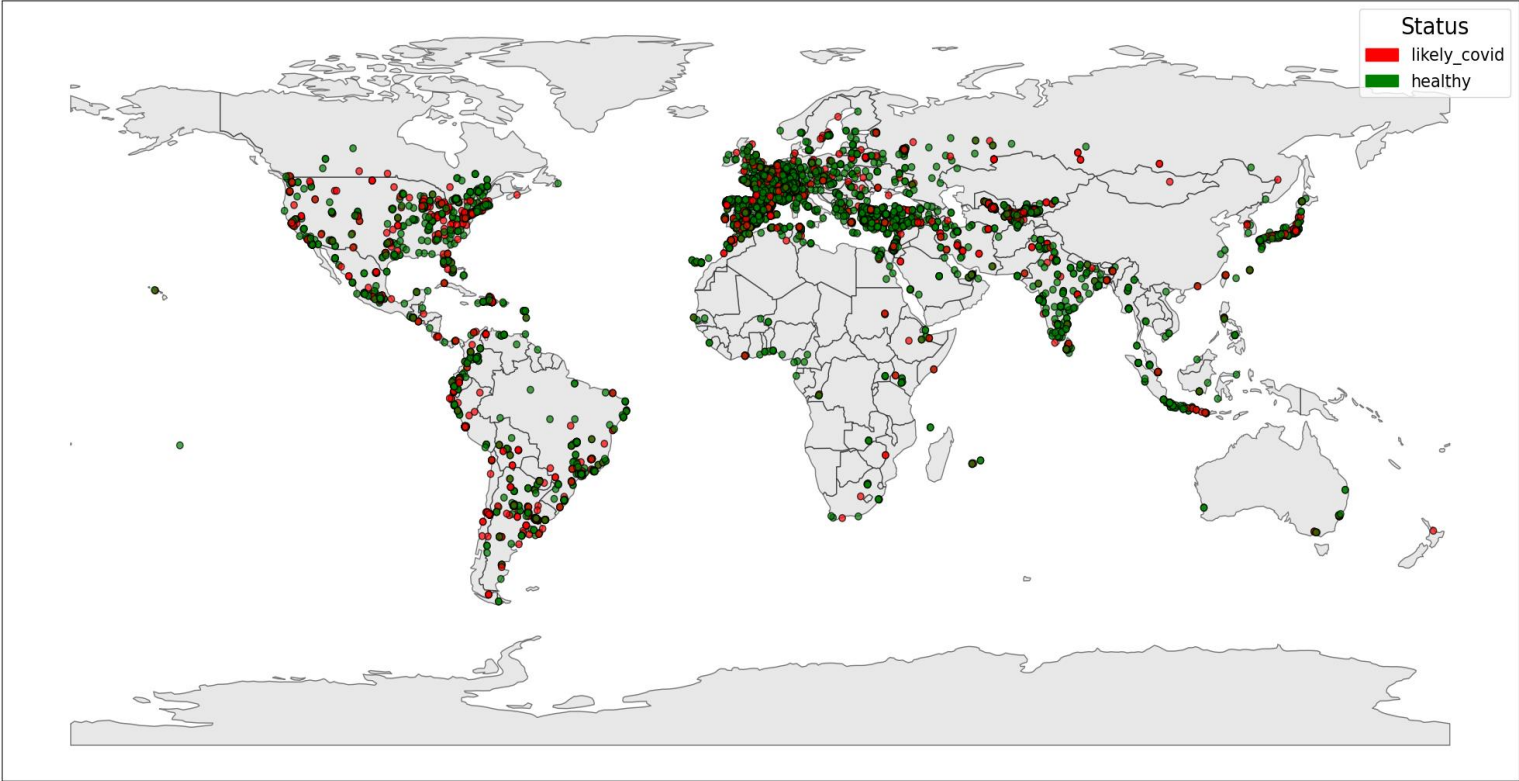
Dataset metadata :

Variable	Possible Values	Description
Datetime	UTC date and time (ISO 8601)	Timestamp of the received recording.
Cough_detected	Float [0, 1]	Probability that the recording contains cough sounds.
SNR	Float [0, ∞)	Estimation of the signal-to-noise ratio of the cough recording.
Latitude	Floating point value	Self-reported latitude with reduced precision.
Longitude	Floating point value	Self-reported longitude with reduced precision.
Age	Integer value	Self-reported age.
Gender	{female, male, other}	Self-reported gender.
Respiratory_condition	{True, False}	Presence of other respiratory conditions (self-reported).
Fever_muscle_pain	{True, False}	Presence of fever or muscle pain (self-reported).
Status	{COVID, symptomatic, healthy}	COVID-19 diagnosis status (self-reported).
Expert_labels_{1,2,3}	JSON dictionary	Expert manual labels (Quality, cough_type,...). -> Only ~1500 patients

Study population

	healthy (N=10628)	likely_covid (N=3561)	Overall (N=14189)
Age (years)			
Mean (SD)	37.1 (14.3)	34.2 (13.1)	36.3 (14.1)
Median [Min, Max]	35.0 [1.00, 99.0]	33.0 [1.00, 97.0]	35.0 [1.00, 99.0]
Missing	530 (5.0%)	159 (4.5%)	689 (4.9%)
Gender			
female	3551 (33.4%)	1511 (42.4%)	5062 (35.7%)
male	7036 (66.2%)	2023 (56.8%)	9059 (63.8%)
other	41 (0.4%)	27 (0.8%)	68 (0.5%)
Respiratory condition			
FALSE	9320 (87.7%)	2511 (70.5%)	11831 (83.4%)
TRUE	1308 (12.3%)	1050 (29.5%)	2358 (16.6%)
Fever or muscle pain			
FALSE	9888 (93.0%)	2684 (75.4%)	12572 (88.6%)
TRUE	740 (7.0%)	877 (24.6%)	1617 (11.4%)

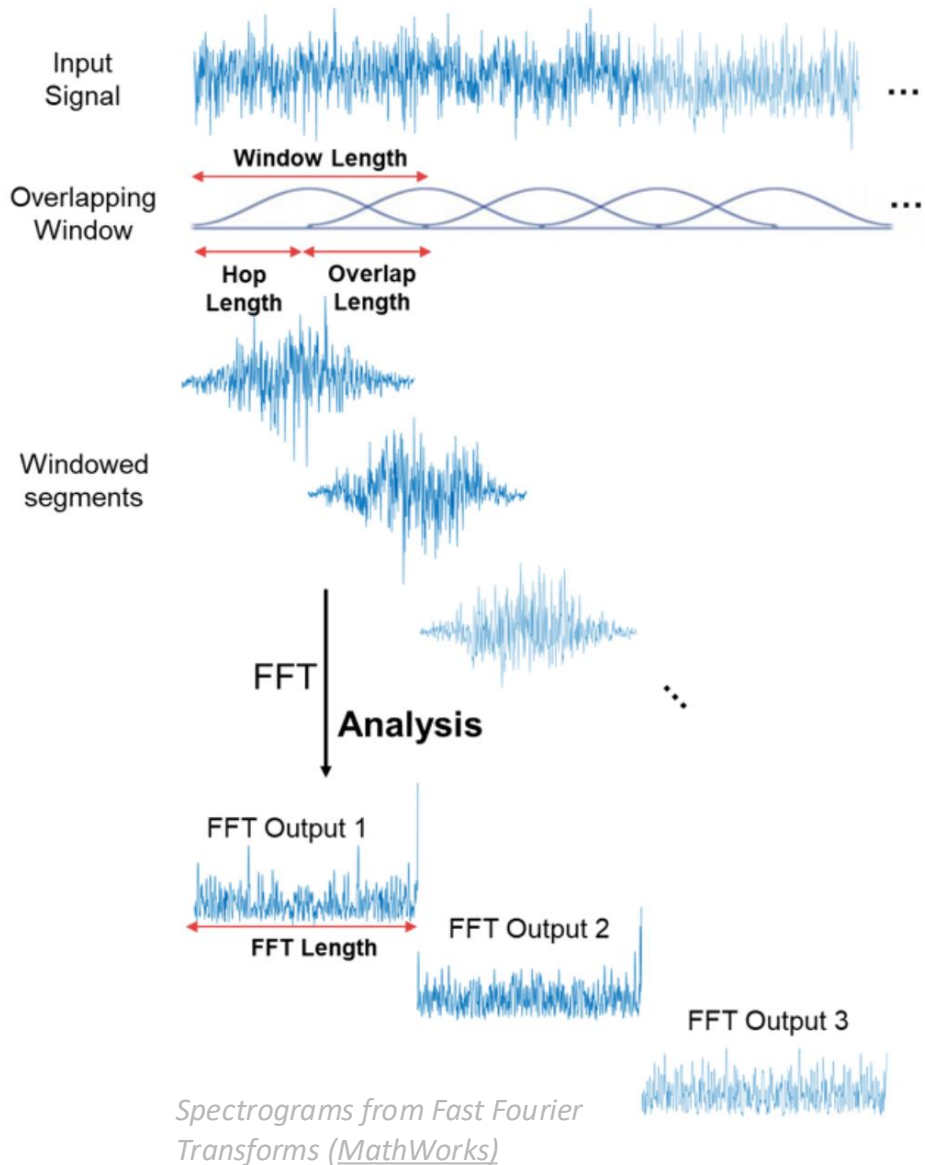
Table 1



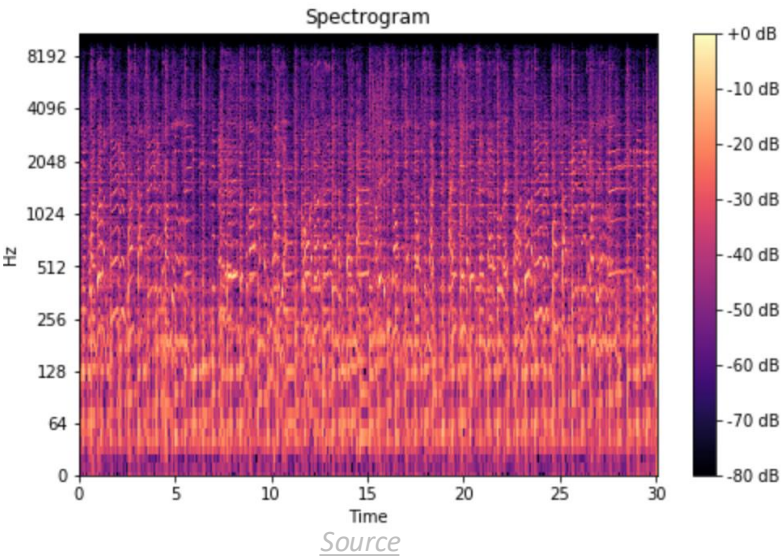
Global view of patient status and localisation*

*Only for people that allowed to share their geolocation information (~5000 NA)

Spectrograms/MelSpectrograms



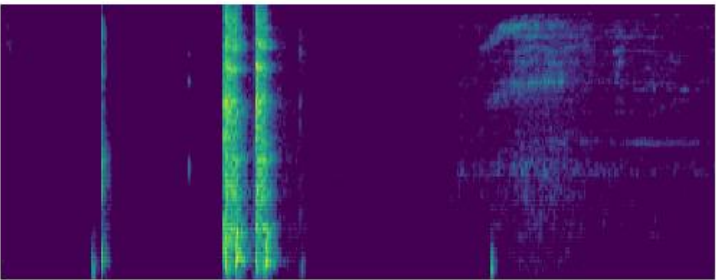
Spectrogram : bunch of FFTs stacked on top of each other to represent a signal’s loudness, or amplitude, as it varies over time at different frequencies.



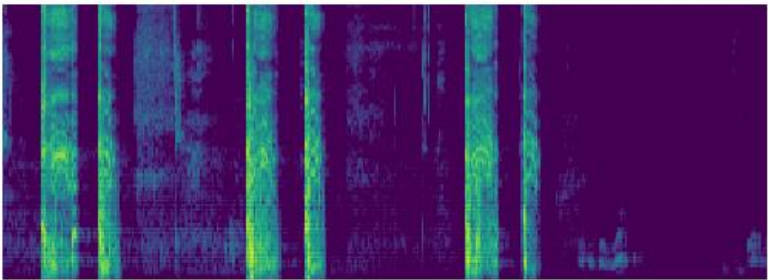
MelSpectrogram: Spectrogram where the frequencies are converted to the mel scale (more similar to human audio perception)

In the dataset

Covid :



Non-Covid:



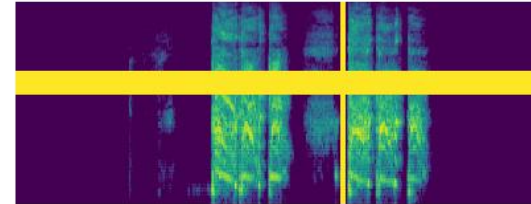
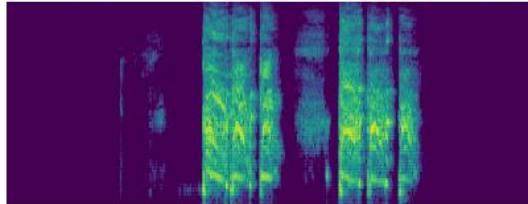
Pitch Shifting & Spectral Data Augmentation

Pitch Shifting

- A technique for adjusting the original sound by increasing or decreasing its pitch
- Used Python library Librosa to shift audio samples down by **four** steps

Spectral Data Augmentation

- *SpecAugment* technique: use mel-spectrograms and a 2-step augmentation method
 1. Frequency masking: Randomly select and mask a range of consecutive mel frequency channels (up to 30 channels)
 2. Time masking: Randomly select and mask a range of consecutive time steps (up to 30 time steps)



References

- [1] Zhang, Boyang et al. "Audio Recognition using Mel Spectrograms and Convolution Neural Networks." (2019).
- [2] Friedrich Wolf-Monheim "Spectral and Rhythm Features for Audio Classification with Deep Convolutional Neural Networks." (2024)
- [3] Quan Zhou et al. "Cough Recognition Based on Mel-Spectrogram and Convolutional Neural Network." (2021)
- [4] Hamdi S, Oussalah M, Moussaoui A, Saidi M. "Attention-based hybrid CNN-LSTM and spectral data augmentation for COVID-19 diagnosis from cough sound" (2023)
- [5] Ali Imran et al. "AI4COVID-19: AI enabled preliminary diagnosis for COVID-19 from cough samples via an app" Informatics in Medicine Unlocked, (2020).
- [6] Bouzid D et al "Comparison of Patients Infected With Delta Versus Omicron COVID-19 Variants Presenting to Paris Emergency Departments : A Retrospective Cohort Study. "(2022)
- [7] Worldometer "COVID-19 Coronavirus Pandemic" (2025)
- [8] Lella, K.K., & Pja, A. "Automatic diagnosis of COVID-19 disease using deep convolutional neural network with multi-feature channel from respiratory sound data: Cough, voice, and breath" (2022).
- [9] Coppock, H., Gaskell, A., Tzirakis, P., Baird, A., Jones, L., & Schuller, B "End-to-end convolutional neural network enables COVID 19 detection from breath and cough audio: a pilot study. BMJ innovations" (2021).
- [10] Orlandic, L., Teijeiro, T. & Atienza, D. The COUGHVID crowdsourcing dataset, a corpus for the study of large-scale cough analysis algorithms. *Sci Data* **8**, 156 (2021).

Appendix – Fourier Transforms and spectrograms

The **Fourier transform** : Decompose a signal into it's individual frequencies and the frequency's amplitude, i.e it converts the signal from the time domain into the frequency domain (possible because every signal can be decomposed into a set of sine and cosine waves that add up to the original signal)

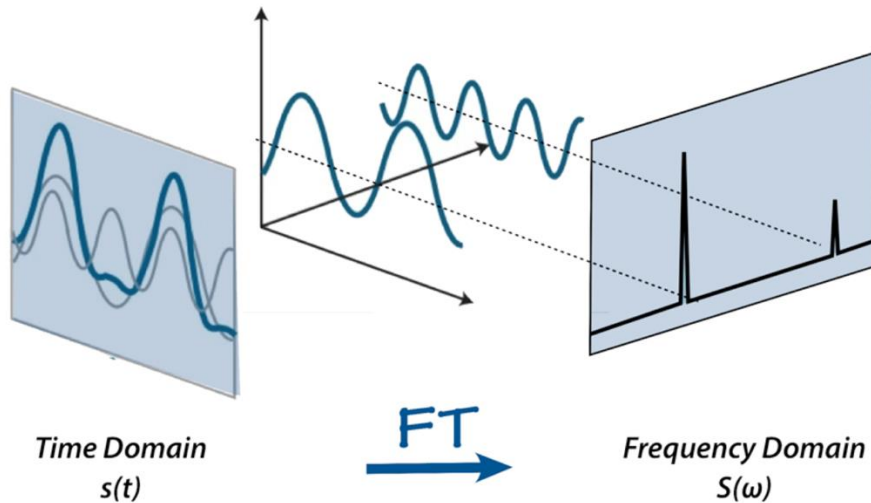


Image from [Aavos International](#)

Mel Scale :

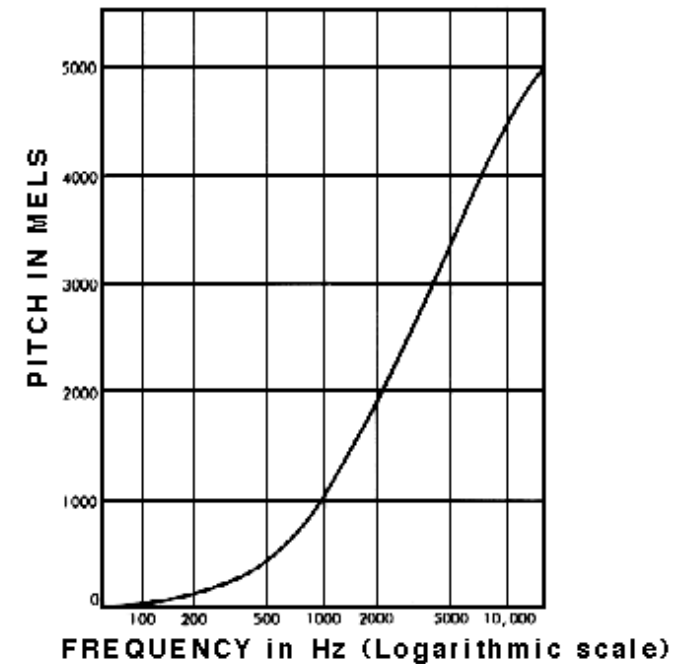


Image from [Simon Fraser University](#)

Appendix – MelSpectograms parameters

Frequency Bands

- f_{min} – minimum frequency
- f_{max} – maximum frequency to display
- n_{mels} – number of frequency bands (ie. Mel bins). This is the height of the Spectrogram

Time Sections

- n_{fft} – window length for each time section
- hop_length – number of samples by which to slide the window at each step (width of the Spectrogram is = Total number of samples / hop_length)